

# Complex Analysis Qualifying Exam Problems

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# August 2018

## Part I

1. Find a conformal map  $f$  from the vertical strip

$$\Omega = \{z \in \mathbb{C} : 3 < \operatorname{Re}(z) < 11\}$$

onto the open unit disk  $\mathbb{D}$ .

*Solution.* The width of the strip is  $11 - 3 = 8$ . First map the strip to a horizontal strip by setting

$$w = \frac{\pi}{8}(z - 3).$$

Since  $3 < \operatorname{Re}(z) < 11$ , we have

$$0 < \operatorname{Re}(w) < \pi.$$

Now define

$$\omega = e^{iw}.$$

Because

$$\arg(\omega) = \operatorname{Re}(w),$$

the map  $\omega = e^{iw}$  sends the strip onto the upper half-plane

$$\{\omega \in \mathbb{C} : \operatorname{Im}(\omega) > 0\}.$$

Finally, the map

$$\phi(\omega) = \frac{\omega - i}{\omega + i}$$

maps the upper half-plane conformally onto the unit disk  $\mathbb{D}$ . Therefore a conformal map from  $\Omega$  onto  $\mathbb{D}$  is

$$f(z) = \frac{e^{i\pi(z-3)/8} - i}{e^{i\pi(z-3)/8} + i}.$$

■

2. Suppose that  $F$  and  $G$  are holomorphic functions on the right half plane

$$H = \{z \in \mathbb{C} : \operatorname{Re}(z) > 0\}.$$

Suppose also that  $F$  and  $G$  have the same zeros with the same multiplicities at each zero. Prove that there is a holomorphic function  $h$  on  $H$  such that

$$F(z) = e^{h(z)}G(z)$$

for all  $z \in H$ .

*Solution.* If  $F \equiv G \equiv 0$ , then we may take  $h \equiv 0$ . Thus assume that  $F$  and  $G$  are not identically zero.

Define

$$q(z) = \frac{F(z)}{G(z)}$$

at points where  $G(z) \neq 0$ . We claim that  $q$  extends to a holomorphic function on all of  $H$ .

Let  $a \in H$  be a zero of  $G$ . Since  $F$  and  $G$  have the same zeros with the same multiplicities, there is some  $m \geq 1$  such that locally

$$F(z) = (z - a)^m \phi(z), \quad G(z) = (z - a)^m \psi(z),$$

where  $\phi$  and  $\psi$  are holomorphic near  $a$ , with

$$\phi(a) \neq 0, \quad \psi(a) \neq 0.$$

Hence near  $a$ ,

$$q(z) = \frac{F(z)}{G(z)} = \frac{\phi(z)}{\psi(z)}.$$

So the singularity of  $q$  at  $a$  is removable. Therefore  $q$  extends holomorphically to all of  $H$ .

Moreover, the extension of  $q$  has no zeros. Indeed, away from the zeros of  $G$ , this is clear since  $F$  and  $G$  have the same zeros. At a common zero  $a$ , we have

$$q(a) = \frac{\phi(a)}{\psi(a)} \neq 0.$$

Now  $H = \{z \in \mathbb{C} : \operatorname{Re}(z) > 0\}$  is convex, hence simply connected. Since  $q$  is holomorphic and never zero on  $H$ , by Corollary 6.17, Chapter IV, there exists a holomorphic function  $h$  on  $H$  such that

$$q(z) = e^{h(z)}.$$

Therefore, for all  $z \in H$ ,

$$F(z) = q(z)G(z) = e^{h(z)}G(z).$$

Thus there is a holomorphic function  $h$  on  $H$  satisfying

$$F(z) = e^{h(z)}G(z)$$

for all  $z \in H$ . ■

**3.** Suppose  $f$  is a holomorphic function on the open unit disc

$$D = \{z : |z| < 1\}.$$

Prove that there is a sequence  $(z_n)_{n=1}^\infty$  in  $D$  such that  $|z_n| \rightarrow 1$  and the set

$$\{f(z_n) : n \geq 1\}$$

is bounded.

*Solution (I).* If  $f \equiv 0$ , then take  $z_n = 1 - \frac{1}{n}$ . Then  $|z_n| \rightarrow 1$  and  $\{f(z_n) : n \geq 1\} = \{0\}$ , which is bounded.

Now assume  $f \not\equiv 0$ . Suppose, toward a contradiction, that no such sequence exists. Then for every  $M > 0$ , there is  $0 < \rho < 1$  such that

$$|f(z)| > M$$

whenever  $\rho < |z| < 1$ . Indeed, otherwise we could find  $z_n \in D$  with

$$|z_n| > 1 - \frac{1}{n} \quad \text{and} \quad |f(z_n)| \leq M,$$

which would give  $|z_n| \rightarrow 1$  and  $\{f(z_n)\}$  bounded.

Thus

$$|f(z)| \rightarrow \infty \quad \text{as } |z| \rightarrow 1.$$

In particular, for some  $0 < R < 1$ ,

$$|f(z)| > 1$$

whenever  $R < |z| < 1$ . Hence all zeros of  $f$  lie in the compact set  $\{|z| \leq R\}$ . Since  $f \not\equiv 0$ , it has only finitely many zeros there. Let those zeros be  $\alpha_1, \dots, \alpha_N$ , with

multiplicities  $m_1, \dots, m_N$ , and define

$$P(z) = \prod_{j=1}^N (z - \alpha_j)^{m_j}.$$

If  $f$  has no zeros, take  $P \equiv 1$ .

Then

$$F(z) = \frac{f(z)}{P(z)}$$

extends to a holomorphic function on  $D$ , and  $F$  has no zeros in  $D$ . Therefore

$$g(z) = \frac{1}{F(z)}$$

is holomorphic on  $D$ .

Since  $P$  is bounded on  $D$ , and  $|f(z)| \rightarrow \infty$  as  $|z| \rightarrow 1$ , we get

$$|F(z)| \rightarrow \infty \quad \text{as } |z| \rightarrow 1.$$

Thus

$$g(z) \rightarrow 0 \quad \text{as } |z| \rightarrow 1.$$

Fix  $z_0 \in D$ . Given  $\varepsilon > 0$ , choose  $\rho < 1$  such that

$$|g(z)| < \varepsilon$$

whenever  $\rho < |z| < 1$ . Choose  $r$  such that

$$\max\{|z_0|, \rho\} < r < 1.$$

By the maximum modulus principle applied on  $\{|z| \leq r\}$ ,

$$|g(z_0)| \leq \max_{|z|=r} |g(z)| < \varepsilon.$$

Since  $\varepsilon > 0$  was arbitrary,  $g(z_0) = 0$ . Hence  $g \equiv 0$  on  $D$ , which is impossible because  $g = 1/F$ .

Therefore our supposition was false. Hence there exists a sequence  $(z_n) \subset D$  such that

$$|z_n| \rightarrow 1$$

and

$$\{f(z_n) : n \geq 1\}$$

is bounded. ■

*Solution (II).* Suppose, toward a contradiction, that no such sequence exists. Then

for every  $M > 0$ , there exists  $0 < \rho < 1$  such that

$$|f(z)| > M$$

whenever  $\rho < |z| < 1$ . Otherwise, for some  $M > 0$ , we could choose  $z_n \in D$  with

$$|z_n| > 1 - \frac{1}{n} \quad \text{and} \quad |f(z_n)| \leq M,$$

which would give the desired sequence.

Thus

$$|f(z)| \rightarrow \infty \quad \text{as } |z| \rightarrow 1.$$

In particular, there exists  $0 < R < 1$  such that

$$|f(z)| > 1$$

whenever  $R < |z| < 1$ . Hence  $f$  has no zeros in the annulus

$$A = \{z : R < |z| < 1\}.$$

Therefore

$$g(z) = \frac{1}{f(z)}$$

is holomorphic on  $A$ . Also,

$$g(z) \rightarrow 0 \quad \text{as } |z| \rightarrow 1.$$

Now expand  $g$  in its Laurent series on the annulus  $A$ :

$$g(z) = \sum_{n=-\infty}^{\infty} c_n z^n.$$

For any  $r$  with  $R < r < 1$ ,

$$c_n = \frac{1}{2\pi i} \int_{|\omega|=r} \frac{g(\omega)}{\omega^{n+1}} d\omega.$$

Hence

$$|c_n| \leq \max_{|\omega|=r} |g(\omega)| r^{-n}.$$

Letting  $r \rightarrow 1^-$ , we have

$$\max_{|\omega|=r} |g(\omega)| \rightarrow 0,$$

and  $r^{-n} \rightarrow 1$ . Therefore

$$c_n = 0$$

for every  $n \in \mathbb{Z}$ . Hence

$$g \equiv 0$$

on  $A$ , which is impossible since  $g = 1/f$ .

Therefore our supposition was false. Thus there exists a sequence  $(z_n) \subset D$  such that

$$|z_n| \rightarrow 1$$

and

$$\{f(z_n) : n \geq 1\}$$

is bounded. ■

**4.** Determine, with proof, all entire functions  $f$  such that for all  $\lambda \in \mathbb{C}$  with  $|\lambda| = 1$  and all  $z \in \mathbb{C}$  the equation

$$f(\lambda z) = \lambda f(z)$$

holds.

*Solution (I).* We claim that the only such functions are

$$f(z) = cz$$

for some constant  $c \in \mathbb{C}$ .

Let  $f$  be entire. Then  $f$  has a power series expansion on all of  $\mathbb{C}$ :

$$f(z) = \sum_{n=0}^{\infty} a_n z^n.$$

For  $|\lambda| = 1$ , we have

$$f(\lambda z) = \sum_{n=0}^{\infty} a_n (\lambda z)^n = \sum_{n=0}^{\infty} a_n \lambda^n z^n.$$

By assumption,

$$f(\lambda z) = \lambda f(z) = \lambda \sum_{n=0}^{\infty} a_n z^n = \sum_{n=0}^{\infty} \lambda a_n z^n.$$

Therefore, for every  $|\lambda| = 1$ ,

$$\sum_{n=0}^{\infty} a_n \lambda^n z^n = \sum_{n=0}^{\infty} \lambda a_n z^n.$$

By uniqueness of power series coefficients,

$$a_n \lambda^n = \lambda a_n$$



for every  $n \geq 0$  and every  $|\lambda| = 1$ . Hence

$$a_n(\lambda^n - \lambda) = 0.$$

If  $n = 1$ , this condition gives no restriction on  $a_1$ . But if  $n \neq 1$ , we can choose  $|\lambda| = 1$  such that

$$\lambda^n \neq \lambda.$$

Thus  $a_n = 0$  for every  $n \neq 1$ . Therefore

$$f(z) = a_1 z.$$

Conversely, every function of the form  $f(z) = cz$  satisfies

$$f(\lambda z) = c\lambda z = \lambda cz = \lambda f(z).$$

Hence the entire functions satisfying the condition are exactly

$$\boxed{f(z) = cz, \quad c \in \mathbb{C}.}$$

■

*Solution (II).* First put  $z = 0$ . Then

$$f(0) = \lambda f(0)$$

for every  $|\lambda| = 1$ . Choosing  $\lambda \neq 1$ , we get

$$f(0) = 0.$$

Hence

$$g(z) = \frac{f(z)}{z}$$

is holomorphic on  $\mathbb{C} \setminus \{0\}$ , and the singularity at 0 is removable. Define

$$g(0) = f'(0).$$

Then  $g$  is entire.

For  $z \neq 0$  and  $|\lambda| = 1$ ,

$$g(\lambda z) = \frac{f(\lambda z)}{\lambda z} = \frac{\lambda f(z)}{\lambda z} = \frac{f(z)}{z} = g(z).$$

Thus, by continuity,

$$g(\lambda z) = g(z)$$

for all  $z \in \mathbb{C}$  and all  $|\lambda| = 1$ .

In particular, if  $|\zeta| = 1$ , then  $\zeta = \lambda \cdot 1$  for some  $|\lambda| = 1$ , so

$$g(\zeta) = g(1).$$

Thus  $g$  is constant on the unit circle.

Write the Taylor expansion of  $g$  at 0:

$$g(z) = \sum_{n=0}^{\infty} a_n z^n.$$

By Cauchy's integral formula for Taylor coefficients,

$$a_n = \frac{1}{2\pi i} \int_{|\zeta|=1} \frac{g(\zeta)}{\zeta^{n+1}} d\zeta.$$

For  $n \geq 1$ , since  $g(\zeta) = g(1)$  on  $|\zeta| = 1$ , we get

$$a_n = \frac{g(1)}{2\pi i} \int_{|\zeta|=1} \frac{1}{\zeta^{n+1}} d\zeta = 0.$$

Therefore all Taylor coefficients of  $g$  except possibly  $a_0$  vanish. Hence  $g$  is constant on  $\mathbb{C}$ . Say

$$g(z) = c.$$

Then

$$f(z) = zg(z) = cz.$$

Conversely, if  $f(z) = cz$ , then

$$f(\lambda z) = c\lambda z = \lambda cz = \lambda f(z).$$

Therefore the desired functions are exactly

$$\boxed{f(z) = cz, \quad c \in \mathbb{C}.}$$

■

**5.** Let  $f$  and  $g$  be entire functions satisfying  $f(0) = g(0) \neq 0$  and

$$|f(z)| \leq |g(z)| \quad \forall z \in \mathbb{C}.$$

Show that  $f = g$ .

*Solution.* Since  $g(0) = f(0) \neq 0$ , there exists  $r > 0$  such that

$$g(z) \neq 0$$

for all  $z \in B(0, r)$ . Define

$$h(z) = \frac{f(z)}{g(z)}$$

on  $B(0, r)$ . Then  $h$  is holomorphic on  $B(0, r)$ . Also, by hypothesis,

$$|h(z)| = \frac{|f(z)|}{|g(z)|} \leq 1$$

for all  $z \in B(0, r)$ . Moreover,

$$h(0) = \frac{f(0)}{g(0)} = 1.$$

Thus

$$|h(0)| = 1.$$

So  $h$  attains its maximum modulus at the interior point 0. By the maximum modulus principle,  $h$  is constant on  $B(0, r)$ . Since  $h(0) = 1$ , we get

$$h \equiv 1$$

on  $B(0, r)$ . Therefore

$$f(z) = g(z)$$

for all  $z \in B(0, r)$ .

Since  $f$  and  $g$  are entire and agree on a nonempty open set, the identity theorem implies that

$$f = g$$

on all of  $\mathbb{C}$ . ■

## Part II

**6.** *Determine all entire functions that satisfy the functional equation*

$$f(3z) = (1 - 3z)f(z).$$

*Express the answer in terms of an infinite product.*

*Solution.* Replacing  $z$  by  $z/3$  in the functional equation gives

$$f(z) = (1 - z)f\left(\frac{z}{3}\right).$$

Iterating this identity  $n$  times, we obtain

$$f(z) = \left( \prod_{k=0}^{n-1} \left( 1 - \frac{z}{3^k} \right) \right) f\left(\frac{z}{3^n}\right).$$

Since  $f$  is entire, it is continuous, and hence

$$\lim_{n \rightarrow \infty} f\left(\frac{z}{3^n}\right) = f(0).$$

Now define

$$P(z) = \prod_{k=0}^{\infty} \left( 1 - \frac{z}{3^k} \right).$$

This product converges uniformly on compact subsets of  $\mathbb{C}$ . Indeed, for  $|z| \leq R$ ,

$$\sum_{k=0}^{\infty} \left| \frac{z}{3^k} \right| \leq R \sum_{k=0}^{\infty} \frac{1}{3^k} = \frac{3R}{2} < \infty.$$

Therefore,  $P$  is entire. Letting  $n \rightarrow \infty$  in the iterated identity gives

$$f(z) = f(0) \prod_{k=0}^{\infty} \left( 1 - \frac{z}{3^k} \right).$$

Thus every entire solution is of the form

$$f(z) = C \prod_{k=0}^{\infty} \left( 1 - \frac{z}{3^k} \right), \quad C \in \mathbb{C}.$$

Conversely, suppose

$$f(z) = C \prod_{k=0}^{\infty} \left( 1 - \frac{z}{3^k} \right).$$

Then

$$\begin{aligned} f(3z) &= C \prod_{k=0}^{\infty} \left( 1 - \frac{3z}{3^k} \right) \\ &= C(1 - 3z) \prod_{k=1}^{\infty} \left( 1 - \frac{z}{3^{k-1}} \right) \\ &= (1 - 3z)f(z). \end{aligned}$$

Hence all entire solutions are

$$f(z) = C \prod_{k=0}^{\infty} \left( 1 - \frac{z}{3^k} \right), \quad C \in \mathbb{C}.$$

■

7. Let  $F(z)$  be holomorphic on a disc  $D(z_0, r)$  so that

$$F(z_0) = F'(z_0) = 0 \quad \text{and} \quad F''(z_0) \neq 0.$$

Show that there are two curves  $\Gamma_1$  and  $\Gamma_2$  passing through  $z_0$  that are orthogonal at  $z_0$  and so that the function  $F$  when restricted to  $\Gamma_1$  is real-valued and has a minimum at  $z_0$ , and when restricted to  $\Gamma_2$  it is also real-valued and has a maximum at  $z_0$ .

*Solution.* Since  $F(z_0) = F'(z_0) = 0$  and  $F''(z_0) \neq 0$ , the function  $F$  has a zero of order exactly two at  $z_0$ . Hence, in a sufficiently small neighborhood of  $z_0$ , we may write

$$F(z) = (z - z_0)^2 g(z),$$

where  $g$  is holomorphic and

$$g(z_0) = \frac{F''(z_0)}{2} \neq 0.$$

After shrinking the neighborhood if necessary,  $g$  has no zeros. Therefore,  $g$  admits a holomorphic square root  $h$ , so that

$$h(z)^2 = g(z).$$

Define

$$\phi(z) = (z - z_0)h(z).$$

Then

$$F(z) = \phi(z)^2.$$

Moreover,

$$\phi(z_0) = 0 \quad \text{and} \quad \phi'(z_0) = h(z_0) \neq 0.$$

Thus, by the holomorphic inverse function theorem,  $\phi$  is biholomorphic from some neighborhood  $U$  of  $z_0$  onto a neighborhood  $V$  of 0. Let

$$\psi = \phi^{-1}.$$

Choose  $\varepsilon > 0$  sufficiently small so that

$$(-\varepsilon, \varepsilon) \cup i(-\varepsilon, \varepsilon) \subseteq V,$$

and define the curves

$$\Gamma_1 = \{\psi(t) : -\varepsilon < t < \varepsilon\}$$

and

$$\Gamma_2 = \{\psi(it) : -\varepsilon < t < \varepsilon\}.$$

Both curves pass through  $z_0$ , since  $\psi(0) = z_0$ .

If  $z = \psi(t) \in \Gamma_1$ , then

$$F(z) = \phi(z)^2 = t^2.$$

Thus  $F$  is real-valued on  $\Gamma_1$  and has a strict minimum at  $z_0$ .

Similarly, if  $z = \psi(it) \in \Gamma_2$ , then

$$F(z) = \phi(z)^2 = (it)^2 = -t^2.$$

Thus  $F$  is real-valued on  $\Gamma_2$  and has a strict maximum at  $z_0$ .

Finally, parametrizing the curves by

$$\gamma_1(t) = \psi(t) \quad \text{and} \quad \gamma_2(t) = \psi(it),$$

we obtain

$$\gamma_1'(0) = \psi'(0) \quad \text{and} \quad \gamma_2'(0) = i\psi'(0).$$

Multiplication by  $i$  rotates a vector through an angle of  $\pi/2$ . Therefore, the tangent vectors to  $\Gamma_1$  and  $\Gamma_2$  at  $z_0$  are orthogonal, and hence the two curves are orthogonal at  $z_0$ . ■

**8.** *Prove that if  $f$  is a non-constant bounded holomorphic function defined on the half plane*

$$\{z : \operatorname{Im} z < 1\},$$

*then there is some point on the real axis where the value of  $f$  is not real.*

*Solution.* Suppose, for contradiction, that

$$f(x) \in \mathbb{R} \quad \text{for every } x \in \mathbb{R}.$$

Define

$$g(z) = \overline{f(\bar{z})}.$$

Since  $f$  is holomorphic on

$$\{z \in \mathbb{C} : \operatorname{Im} z < 1\},$$

the function  $g$  is holomorphic on

$$\{z \in \mathbb{C} : \operatorname{Im} z > -1\}.$$

Both  $f$  and  $g$  are holomorphic on the strip

$$S = \{z \in \mathbb{C} : -1 < \operatorname{Im} z < 1\}.$$

For every  $x \in \mathbb{R}$ , we have

$$g(x) = \overline{f(x)} = f(x),$$

because  $f(x)$  is real. Since the real axis has accumulation points in  $S$ , the identity theorem implies that

$$g(z) = f(z) \quad \text{for every } z \in S.$$

Therefore, the function

$$F(z) = \begin{cases} f(z), & \text{Im } z < 1, \\ \overline{f(\bar{z})}, & \text{Im } z > -1 \end{cases}$$

is well-defined and entire.

Since  $f$  is bounded, there exists  $M > 0$  such that

$$|f(z)| \leq M \quad \text{whenever } \text{Im } z < 1.$$

If  $\text{Im } z > -1$ , then

$$\text{Im}(\bar{z}) < 1,$$

so

$$|F(z)| = \left| \overline{f(\bar{z})} \right| = |f(\bar{z})| \leq M.$$

Thus  $F$  is a bounded entire function. By Liouville's theorem,  $F$  is constant. Hence  $f$  is constant, contradicting the hypothesis. Therefore, there exists some  $x \in \mathbb{R}$  such that

$$f(x) \notin \mathbb{R}.$$

■

**9.** Show that there exists a sequence of polynomials  $(P_n)$  so that

$$\lim_{n \rightarrow \infty} P_n(z) = \begin{cases} 1, & \text{if } \text{Re}(z) > 0, \\ 0, & \text{if } \text{Re}(z) = 0, \\ -1, & \text{if } \text{Re}(z) < 0. \end{cases}$$

Also, explain why  $(P_n)$  can't be bounded on compact subsets of  $\mathbb{C}$ .

*Solution.* For each  $n \in \mathbb{N}$ , define

$$K_n^+ = \left\{ x + iy : \frac{1}{n} \leq x \leq n, |y| \leq n \right\},$$

$$K_n^- = \left\{ x + iy : -n \leq x \leq -\frac{1}{n}, |y| \leq n \right\},$$

and

$$K_n^0 = \{iy : |y| \leq n\}.$$

Let

$$K_n = K_n^+ \cup K_n^- \cup K_n^0.$$

The set  $K_n$  is compact, and its complement is connected. Define  $g_n : K_n \rightarrow \mathbb{C}$  by

$$g_n(z) = \begin{cases} 1, & z \in K_n^+, \\ -1, & z \in K_n^-, \\ 0, & z \in K_n^0. \end{cases}$$

The three sets  $K_n^+$ ,  $K_n^-$ , and  $K_n^0$  are pairwise disjoint, so  $g_n$  is continuous on  $K_n$ . Moreover,  $g_n$  is holomorphic on the interior of  $K_n$ .

By Mergelyan's theorem, there exists a polynomial  $P_n$  such that

$$\sup_{z \in K_n} |P_n(z) - g_n(z)| < \frac{1}{n}.$$

Let  $z = x + iy \in \mathbb{C}$ . If  $x > 0$ , then for all sufficiently large  $n$ ,

$$\frac{1}{n} \leq x \leq n \quad \text{and} \quad |y| \leq n.$$

Thus  $z \in K_n^+$ , and hence

$$|P_n(z) - 1| < \frac{1}{n}.$$

Therefore,

$$\lim_{n \rightarrow \infty} P_n(z) = 1.$$

If  $x < 0$ , then for all sufficiently large  $n$ , we have  $z \in K_n^-$ , so

$$|P_n(z) + 1| < \frac{1}{n}.$$

Therefore,

$$\lim_{n \rightarrow \infty} P_n(z) = -1.$$

Finally, if  $x = 0$ , then  $z = iy$ , and for all sufficiently large  $n$ , we have  $z \in K_n^0$ . Hence

$$|P_n(z)| < \frac{1}{n},$$

and therefore

$$\lim_{n \rightarrow \infty} P_n(z) = 0.$$



Consequently,

$$\lim_{n \rightarrow \infty} P_n(z) = \begin{cases} 1, & \operatorname{Re}(z) > 0, \\ 0, & \operatorname{Re}(z) = 0, \\ -1, & \operatorname{Re}(z) < 0. \end{cases}$$

It remains to show that  $(P_n)$  cannot be bounded on compact subsets of  $\mathbb{C}$ . Suppose, for contradiction, that it were bounded on compact subsets. Thus, for every compact set  $K \subseteq \mathbb{C}$ , there would exist a constant  $M_K$  such that

$$|P_n(z)| \leq M_K \quad \text{for every } z \in K \text{ and every } n.$$

The family  $\{P_n\}$  would then be locally bounded. By Montel's theorem, there would exist a subsequence  $(P_{n_k})$  converging uniformly on compact subsets of  $\mathbb{C}$  to an entire function  $P$ .

On the other hand, every subsequence of  $(P_n)$  has the same pointwise limit

$$G(z) = \begin{cases} 1, & \operatorname{Re}(z) > 0, \\ 0, & \operatorname{Re}(z) = 0, \\ -1, & \operatorname{Re}(z) < 0. \end{cases}$$

Therefore,  $P = G$  on  $\mathbb{C}$ . This is impossible because  $P$  is entire and hence continuous, whereas  $G$  is discontinuous at every point of the imaginary axis.

Thus,  $(P_n)$  cannot be bounded on compact subsets of  $\mathbb{C}$ . ■

**10.** Let  $\mathfrak{F}$  be the collection of holomorphic functions on the open unit disk

$$D = \{z : |z| < 1\}$$

satisfying

$$\iint_D |f| \, dx \, dy \leq 1.$$

Prove or disprove that  $\mathfrak{F}$  is a normal family.

# August 2020

## Part I

1. Let  $f$  be an entire function. Show that  $f$  is a constant function if there exist  $M > 0$  and  $R > 0$  such that

$$|f(z)| \leq M|z|^{1/2} \quad \text{whenever } |z| > R.$$

*Solution.* Expand  $f$  in its Taylor series about 0:

$$f(z) = \sum_{n=0}^{\infty} a_n z^n.$$

Let  $r > R$ . By Cauchy's estimate,

$$|a_n| \leq \frac{\max_{|z|=r} |f(z)|}{r^n}.$$

Since

$$|f(z)| \leq M|z|^{1/2}$$

whenever  $|z| > R$ , we have

$$\max_{|z|=r} |f(z)| \leq Mr^{1/2}.$$

Therefore,

$$|a_n| \leq Mr^{1/2-n}.$$

For every  $n \geq 1$ , we have  $1/2 - n < 0$ . Letting  $r \rightarrow \infty$ , we obtain

$$|a_n| = 0.$$

Thus  $a_n = 0$  for every  $n \geq 1$ . Hence

$$f(z) = a_0$$

for all  $z \in \mathbb{C}$ , and therefore  $f$  is constant. ■

2. Fix  $w = re^{i\theta} \neq 0$  and let  $\gamma$  be a piecewise smooth curve in  $\mathbb{C} \setminus \{0\}$  from 1 to  $w$ .

Show that there is an integer  $k$  such that

$$\int_{\gamma} \frac{1}{z} dz = \log r + i\theta + 2\pi ik.$$

*Solution.* Let  $\sigma$  be the path from 1 to  $r$  along the positive real axis, followed by the circular arc from  $r$  to  $w = re^{i\theta}$ . More precisely, the first part is given by

$$\sigma_1(t) = t, \quad 1 \leq t \leq r,$$

and the second part is given by

$$\sigma_2(t) = re^{it}, \quad 0 \leq t \leq \theta.$$

Then

$$\int_{\sigma_1} \frac{1}{z} dz = \int_1^r \frac{1}{t} dt = \log r,$$

and

$$\int_{\sigma_2} \frac{1}{z} dz = \int_0^\theta \frac{1}{re^{it}} ire^{it} dt = i\theta.$$

Therefore,

$$\int_{\sigma} \frac{1}{z} dz = \log r + i\theta.$$

Now let

$$\Gamma = \gamma * (-\sigma),$$

where  $-\sigma$  denotes the reverse of  $\sigma$ . Then  $\Gamma$  is a closed curve in  $\mathbb{C} \setminus \{0\}$ . Hence, by the winding number formula,

$$\int_{\Gamma} \frac{1}{z} dz = 2\pi ik$$

for some  $k \in \mathbb{Z}$ .

Since

$$\int_{\Gamma} \frac{1}{z} dz = \int_{\gamma} \frac{1}{z} dz - \int_{\sigma} \frac{1}{z} dz,$$

we obtain

$$\int_{\gamma} \frac{1}{z} dz - (\log r + i\theta) = 2\pi ik.$$

Therefore,

$$\int_{\gamma} \frac{1}{z} dz = \log r + i\theta + 2\pi ik$$

for some  $k \in \mathbb{Z}$ . ■

3. (a) Let  $a > 1$ . Show that the equation

$$ze^{a-z} = 1$$

has precisely one root in the closed unit disk  $\{z : |z| \leq 1\}$ .

(b) Show that the root in part (a) is positive.

*Solution.* (a) Define

$$F(z) = ze^{a-z} - 1 \quad \text{and} \quad G(z) = ze^{a-z}.$$

On the unit circle  $|z| = 1$ , we have

$$|G(z)| = |z|e^{a-\operatorname{Re} z} = e^{a-\operatorname{Re} z}.$$

Since  $\operatorname{Re} z \leq 1$  and  $a > 1$ ,

$$|G(z)| \geq e^{a-1} > 1 = |F(z) - G(z)|.$$

Therefore, by Rouché's theorem,  $F$  and  $G$  have the same number of zeros in the open unit disk.

Since  $e^{a-z} \neq 0$  for every  $z \in \mathbb{C}$ , the function

$$G(z) = ze^{a-z}$$

has exactly one zero, namely  $z = 0$ , and this zero is simple. Thus  $F$  has exactly one zero in the open unit disk.

Moreover, the strict inequality above shows that  $F$  has no zeros on  $|z| = 1$ . Hence the equation

$$ze^{a-z} = 1$$

has precisely one root in the closed unit disk

$$\{z \in \mathbb{C} : |z| \leq 1\}.$$

(b) Consider the real-valued continuous function

$$h(x) = xe^{a-x} - 1, \quad 0 \leq x \leq 1.$$

We have

$$h(0) = -1 < 0$$

and

$$h(1) = e^{a-1} - 1 > 0,$$

since  $a > 1$ . By the intermediate value theorem, there exists  $x_0 \in (0, 1)$  such that

$$h(x_0) = 0.$$

Thus

$$x_0 e^{a-x_0} = 1.$$

Therefore  $x_0$  is a positive root lying in the unit disk. By the uniqueness established in part (a), it is the unique root in the closed unit disk. Hence that root is positive. ■

4. Using the maximum modulus principle, prove the Fundamental Theorem of Algebra.

5. Let  $f$  be an entire function, taking real values on the real axis, and imaginary values on the imaginary axis. Show that  $f(z)$  is an odd function; that is,

$$f(-z) = -f(z)$$

for all  $z$ .

## Part II

6. Given entire functions  $f, g$  without common zeros, prove that there exist entire functions  $A, B$  such that

$$Af + Bg = 1.$$

*Hint: Use the Mittag-Leffler Theorem.*

*Solution.* Assume first that  $f \not\equiv 0$ , and let  $\{a_n\}$  be the zeros of  $f$ , listed without repetition. Since  $f$  and  $g$  have no common zeros, we have

$$g(a_n) \neq 0$$

for every  $n$ .

For each  $n$ , let  $P_n$  be the principal part at  $a_n$  of the Laurent expansion of the meromorphic function

$$\frac{1}{fg}.$$

By the Mittag-Leffler theorem, there exists a meromorphic function  $h$  on  $\mathbb{C}$  whose only possible poles are the points  $a_n$ , and whose principal part at  $a_n$  is  $P_n$ .

Define

$$B = fh$$

and

$$A = \frac{1}{f} - gh.$$

We claim that  $A$  and  $B$  are entire. Suppose that  $a_n$  is a zero of  $f$  of multiplicity  $m$ . Since  $g(a_n) \neq 0$ , the function  $1/(fg)$  has a pole of order  $m$  at  $a_n$ . Hence  $h$  has a pole of order at most  $m$  at  $a_n$ , so the singularity of  $fh$  at  $a_n$  is removable. Therefore,  $B = fh$  is entire.

Moreover,  $h$  and  $1/(fg)$  have the same principal part at each  $a_n$ . Thus

$$\frac{1}{fg} - h$$

is holomorphic at every zero of  $f$ . Since

$$A = \frac{1}{f} - gh = g \left( \frac{1}{fg} - h \right),$$

the function  $A$  is holomorphic at every zero of  $f$ . Away from the zeros of  $f$ , it is clearly holomorphic. Hence  $A$  is entire.

Finally,

$$Af + Bg = \left( \frac{1}{f} - gh \right) f + (fh)g = 1 - fgh + fgh = 1.$$

Therefore, there exist entire functions  $A$  and  $B$  such that

$$Af + Bg = 1.$$

If  $f \equiv 0$ , then  $g$  has no zeros, so  $1/g$  is entire. In that case, take

$$A = 0, \quad B = \frac{1}{g}.$$

■

**7.** If  $f$  is a nonconstant entire function, show that

$$(f^3 + f^2)(\mathbb{C}) = \mathbb{C}.$$

**8.** Suppose that  $G$  is a region in  $\mathbb{C}$ ,  $\{f_n\}$  is a sequence in  $H(G)$ ,  $f$  is a nonconstant function, and  $f_n \rightarrow f$  in  $H(G)$ . Let  $a \in G$  and  $b = f(a)$ . Show that there is a sequence  $\{a_n\}$  in  $G$  such that

$$(i) \ a = \lim a_n,$$

(ii)  $f_n(a_n) = b$  for sufficiently large  $n$ .

**9.** Let

$$G = \{z : \operatorname{Re}(z) > 0\},$$

and  $f : G \rightarrow G$  be a one-one analytic function from  $G$  into  $G$ . Suppose  $f(a) = a$  for some positive real number  $a$ . Show that

$$|f'(a)| \leq 1.$$

**10.** Let  $0 < R_1 < R_2$  and  $\alpha \in \mathbb{R}$ , and let  $G$  be the annulus centered at the origin with radii  $R_1$  and  $R_2$ . Let

$$z^\alpha = \exp(\alpha \log z),$$

where  $\log z$  is a branch of the logarithm. Prove that  $z^\alpha$  is analytic in  $G$  if and only if  $\alpha$  is an integer.

# August 2024

## Part I

1. Let  $G$  and  $\Omega$  be two regions in  $\mathbb{C}$ , and  $h : G \rightarrow \mathbb{C}$  be a one-one analytic function, and  $f : G \rightarrow \mathbb{C}$  be a continuous function with  $f(G) \subset \Omega$ , and  $g : \Omega \rightarrow \mathbb{C}$  be an analytic function with  $g'(w) \neq 0$  for any  $w \in \Omega$ . Show that  $f$  is analytic if

$$h(z) = g(f(z)), \quad \text{for all } z \in G.$$

2. Let  $D$  be the open unit disk. Show that no analytic function  $f : D \rightarrow \mathbb{C}$  has the property that

$$|f(z_n)| \rightarrow \infty, \quad \text{whenever } |z_n| \rightarrow 1.$$

3. How many roots does the polynomial

$$z^9 + z^5 - 8z^3 + 2z + 1$$

have between the circles  $\{|z| = 1\}$  and  $\{|z| = 2\}$ ?

4. Prove Liouville's Theorem by considering the following integral

$$\int_{C_R} \frac{f(z)}{(z-a)(z-b)} dz,$$

for a bounded entire function  $f(z)$ , where  $C_R = Re^{it}$  with  $t \in [0, 2\pi]$ , and  $a \neq b$  with  $|a| < R$  and  $|b| < R$ .

5. Let  $G$  be a region, and

$$G^* = \{z : \bar{z} \in G\}.$$

If  $f : G \rightarrow \mathbb{C}$  is analytic, prove that  $f^* : G^* \rightarrow \mathbb{C}$  defined by

$$f^*(z) = \overline{f(\bar{z})}$$

is also analytic.



## Part II

6. Prove that a nonconstant harmonic function on a region is an open map.
7. Let  $G$  be a region and  $f, f_1, f_2, \dots$  be elements of  $H(G)$ . Show that  $f_n \rightarrow f$  in  $H(G)$  if and only if for each closed rectifiable curve  $\gamma$  in  $G$ ,  $f_n \rightarrow f$  uniformly on  $\gamma$ .
8. Let  $G$  be a region and suppose  $\mathcal{F}$  is normal in  $H(G)$  and  $\Omega$  is open in  $\mathbb{C}$  such that  $f(G) \subset \Omega$  for every  $f \in \mathcal{F}$ . Show that if  $g$  is analytic on  $\Omega$  and is bounded on bounded sets then

$$\{g \circ f : f \in \mathcal{F}\}$$

is normal.

9. Show there is an analytic function  $f$  on

$$D = \{z : |z| < 1\}$$

which is not analytic on any region  $G$  which properly contains  $D$ .

10. Find an entire function  $f$  such that for every integer  $n \geq 1$ ,  $f$  has  $n$  as a zero of order  $n$ . Find the most elementary solution possible.

# January 2025

## Part I

1. *Prove the Fundamental Theorem of Algebra by completing the following two steps for a nonconstant polynomial*

$$p(z) = a_0 + a_1z + \cdots + a_nz^n,$$

where  $a_n \neq 0$  and  $n \geq 1$ .

(a) *Show that there is a nonzero polynomial  $q(z)$  such that*

$$\frac{1}{zp(z)} = \frac{1}{zp(0)} + \frac{q(z)}{p(z)},$$

for  $z \neq 0$  and  $p(z) \neq 0$ .

(b) *By means of integration over the circle  $C_R = Re^{it}$  with  $t \in [0, 2\pi]$ , show that  $p(z_0) = 0$  for some point  $z_0$  in  $\mathbb{C}$ .*

2. *Show that the equation*

$$e^z = cz,$$

where  $c$  is a constant with  $|c| > e$ , has exactly one solution in the set

$$\{z \in \mathbb{C} : \operatorname{Re} z < 1\}.$$

3. *Show that if  $f : \mathbb{C} \rightarrow \mathbb{C}$  is a continuous function such that  $f$  is analytic off  $[-1, 1]$ , then  $f$  is an entire function.*

4. *Using Liouville's Theorem, prove the Fundamental Theorem of Algebra.*

5. *Let  $G$  be a region and  $z_0 \in G$ . Suppose  $f : G \rightarrow \mathbb{C}$  is analytic and  $f'(z_0) \neq 0$ . Show that*

$$\int_{\gamma} \frac{1}{f(z) - f(z_0)} dz = \frac{2\pi i}{f'(z_0)}$$

if  $\gamma = \gamma(t) = z_0 + re^{it}$  for  $t \in [0, 2\pi]$  and  $r > 0$  is small enough.

## Part II

6. Let  $G$  and  $\Omega$  be domains in the complex plane. If  $f : G \rightarrow \Omega$  is one-to-one and analytic and  $\varphi : \Omega \rightarrow \mathbb{R}$  is subharmonic, show that  $\varphi \circ f$  is subharmonic.

7. Let  $G$  be a domain in the complex plane and  $u : G \rightarrow \mathbb{R}$  be a nonconstant harmonic function and let

$$A = \{z \in G : u_x(z) = u_y(z) = 0\}.$$

Show that  $A$  does not have any limit points in  $G$ .

*Solution.* Define

$$f(z) = u_x(z) - i u_y(z).$$

Since  $u$  is harmonic on  $G$ , we have

$$u_{xx} + u_{yy} = 0.$$

Also, harmonic functions are smooth, so

$$u_{xy} = u_{yx}.$$

Thus, the real and imaginary parts of  $f$ , namely  $u_x$  and  $-u_y$ , satisfy the Cauchy–Riemann equations:

$$(u_x)_x = (-u_y)_y$$

and

$$(u_x)_y = -(-u_y)_x.$$

Therefore,  $f$  is holomorphic on  $G$ . Moreover,

$$A = \{z \in G : u_x(z) = u_y(z) = 0\} = \{z \in G : f(z) = 0\}.$$

Hence  $A$  is the zero set of the holomorphic function  $f$ . Suppose that  $A$  has a limit point in  $G$ . By the identity theorem,

$$f \equiv 0$$

on  $G$ . Consequently,

$$u_x \equiv 0 \quad \text{and} \quad u_y \equiv 0$$

on  $G$ . Since  $G$  is connected, this implies that  $u$  is constant on  $G$ , contradicting the assumption that  $u$  is nonconstant. Therefore,  $A$  has no limit points in  $G$ . ■

8. Find an entire function  $f$  such that

$$f(ne^{2k\pi i/n}) = 0$$

for all positive integers  $n$  and all integers  $0 \leq k \leq n-1$ . Give the most elementary example possible.

*Solution.* For each positive integer  $n$ , the points

$$ne^{2k\pi i/n}, \quad 0 \leq k \leq n-1,$$

are precisely the roots of the polynomial  $z^n - n^n$ . Thus, consider

$$f(z) = \prod_{n=1}^{\infty} \left(1 - \frac{z^n}{n^n}\right).$$

We first show that this product defines an entire function. Fix  $R > 0$ . For  $|z| \leq R$ , we have

$$\left|\frac{z^n}{n^n}\right| \leq \left(\frac{R}{n}\right)^n.$$

If  $n \geq 2R$ , then

$$\left(\frac{R}{n}\right)^n \leq 2^{-n}.$$

Consequently,

$$\sum_{n=1}^{\infty} \sup_{|z| \leq R} \left|\frac{z^n}{n^n}\right| \leq \sum_{n=1}^{\infty} \left(\frac{R}{n}\right)^n < \infty.$$

Therefore, the infinite product

$$\prod_{n=1}^{\infty} \left(1 - \frac{z^n}{n^n}\right)$$

converges uniformly on every compact subset of  $\mathbb{C}$ . Since its partial products are polynomials,  $f$  is entire. Moreover,  $f(0) = 1$ , so  $f$  is not identically zero. Now let  $n$  be a positive integer and let  $0 \leq k \leq n-1$ . Then

$$(ne^{2k\pi i/n})^n = n^n e^{2k\pi i} = n^n.$$

Hence the  $n$ -th factor in the product vanishes:

$$1 - \frac{(ne^{2k\pi i/n})^n}{n^n} = 0.$$

It follows that

$$f(ne^{2k\pi i/n}) = 0.$$

Thus, an elementary example is

$$f(z) = \prod_{n=1}^{\infty} \left(1 - \frac{z^n}{n^n}\right).$$

■

**9.** Let  $G$  and  $\Omega$  be two domains in the complex plane. Suppose  $\{f_n\}$  is a sequence in  $C(G, \Omega)$  which converges to  $f$  and  $\{z_n\}$  is a sequence in  $G$  which converges to a point  $z$  in  $G$ . Show

$$\lim_{n \rightarrow \infty} f_n(z_n) = f(z).$$

**10.** Show that if  $\mathcal{F} \subset H(G)$  is normal then

$$\mathcal{F}' = \{f' : f \in \mathcal{F}\}$$

is also normal.